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Ion Mobility Suite (IMoS) 2: A General Computational Framework for Solving the Generalized Boltzmann Equation with Rotational Transport for Polyatomic Ions in Arbitrary Fields

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ABSTRACT: Ion mobility spectrometry (IMS) depends on accurate descriptions of ion−neutral collision dynamics to relate measured mobilities to molecular structure; however, existing approaches based on collision integral quadratures or rigid-body trajectory methods are largely restricted to low-field, elastic, and near-equilibrium assumptions. Here we present IMoS 2, a trajectory-resolved, first-principles computational framework that directly realizes the generalized Boltzmann equation with Curtiss-type rotational transport and a Wang−Chang−Uhlenbeck (WCU) collision operator for polyatomic ions in arbitrary electric fields. Rather than evaluating high-dimensional collision integrals, the method propagates ions within an explicit neutral gas environment using Monte Carlo molecular dynamics, explicitly coupling translational and rotational motion and permitting multiple simultaneous collisions. This strategy provides, in a single simulation, time-resolved trajectories, translational and angular velocity distribution functions, and transport observables such as drift velocity and mobility. In the low-field limit, IMoS 2 quantitatively reproduces Mason−Schamp mobilities and agrees with IMoS 1 and conventional quadrature methods in He and N_2. Temperature-dependent trends match available experiments, and simulations under arbitrary field profiles agree with two-temperature and collision-integral predictions for light monatomic gases, while deviations at higher fields in heavier gases are captured and attributed to inelastic energy transfer into internal degrees of freedom. To our knowledge, IMoS 2 provides the first direct numerical realization of the Curtiss−Wang−Chang−Uhlenbeck kinetic equation for polyatomic ions, establishing a general platform that recovers classical transport theory where valid and extends IMS modeling to nonequilibrium, field-dependent regimes beyond traditional quadrature approaches.

■ INTRODUCTION

Ion mobility spectrometry (IMS) has established itself as a central technique in gas-phase analysis, enabling the separation of ions through a buffer gas under the influence of an electric field.[1−4] Its coupling with mass spectrometry (MS) has transformed structural biology, materials science, and environmental chemistry, providing insights into ion conformations, complexes, and aerosols.[5−9] The quantitative interpretation of IMS experiments relies on connecting mobility to structure through collision cross sections (CCS), most commonly expressed in the low-field limit by the Mason−Schamp equation and quadratures therein.[10−14] Accurate evaluation of the collision integrals in the expanded version of the quadrature, remains one of the most enduring theoretical challenges in the field.[15−19]

Over the past three decades, several strategies have been developed to calculate mobility and CCS from structural information. Analytical approximations such as the Projection Approximation (PA) and its refinements (such as IMPACT),[20] offer computational speed but lack physical rigor when potential interactions or nonconvex geometries dominate even when potential corrections can be added. The introduction of MOBCAL represented a watershed moment, as it implemented exact hard-sphere scattering (EHSS) and the trajectory method (TM) using Lennard−Jones (L−J) and ion−induced dipole potentials, achieving CCS predictions typically within a few percent of experimental values for small to medium ions.[21,22] Since then, numerous variants have emerged, including Mobcal-MPI, Collidoscope, and HPCCS, aimed primarily at accelerating MOBCAL through parallelization or modifications to interaction potentials.[23−26] In parallel, shape-based approximations such as the Projection Superposition Approximation (PSA) have been used to extend CCS prediction to very large biomolecules.[27]

Despite these successes, existing approaches have fundamental limitations. Most trajectory methods treat ions as rigid structures, neglecting rotational or vibrational degrees of freedom that can influence mobility, particularly for polyatomic ions or at high fields. Moreover, the scattering physics embedded in MOBCAL and its derivatives are largely elastic and specular, limiting their ability to capture diffuse or inelastic collisions that become significant for larger particles and heavier gases. These shortcomings motivated the development of the Ion Mobility

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Software (IMoS) suite, which introduced a momentum transfer approach based on Epstein and Rosner's work to calculate mobility and its tensorial nature.[28−35] Despite its different approach, IMoS has been benchmarked against MOBCAL, reproducing results within 1−2% when using the same potentials, but at a fraction of the computational cost.[36] More importantly, it enabled calculations in scattering regimes inaccessible to MOBCAL, broadening the theoretical toolkit available to the IMS community.[37] It was also pioneering in expanding MOBCAL calculations for polyatomic ions from low-fields to arbitrary fields using the two-temperature theory approach and offering ways to calculate dispersion plots from DMS and FAIMS systems.[38,39] The ability to calculate high field mobilities through higher order CCS and matrix elements is superb in itself but incredibly arduous and intractable, which makes further theoretical progress extremely difficult.. It also exposes issues inaccessible to conventional theory such as inelastic collisions, internal heating, relaxation effects and many others.[40]

The next logical step is therefore to move beyond static Mason-Schamp quadrature calculations to a more dynamical simulation of ions in drift.[7,41−49] This also naturally moves the theoretical framework from the Boltzmann-type treatment to the Wang Chang−Uhlenbeck−de Boer (WUB/WCUB) kinetic equation, which extends the translational argument to include rotational and vibrational energy exchanges.[50−52] Although the WUB formalism is analytically intractable for complex ions, Monte Carlo approximations of its physics offer a practical path forward, in particular through a momentum transfer perspective. In a dynamical simulation, the ions will retain memory of previous collisions, translational and rotational effects and internal temperatures.[53−56] This is key to the algorithm, as it enables the application of spatially and temporally varying electric fields and allows the influence of an ion's dipole moment, center of mass (CoM), and moment of inertia (MoI) to be systematically evaluated.[57−60] Arbitrary fields may be employed, and mixtures of gases can be studied.

Here we present IMoS 2, a publicly released, fully quantitative simulation platform designed to implement these theoretical advances. IMoS 2 explicitly couples translational and rotational degrees of freedom, enabling the study of ion mobility shifts arising from CoM and MoI variations, rotational heating, and alignment phenomena for an arbitrarily varying field. The algorithm is quantitative, delivering mobilities consistent with experimental values and established mobility software, while also revealing effects beyond the reach of traditional methods. The software is parallelized to handle billions of collisions efficiently and is distributed with a comprehensive manual and worked examples to facilitate adoption by both theorists and experimentalists. While the current release focuses on translation−rotation coupling, its architecture is designed to accommodate vibrational degrees of freedom, paving the way for more complete treatments of inelastic collisions. And while its focus here is on mobility, the algorithm fully solves the Boltzmann (WUB) equation for a polyatomic ion of negligible concentration subject to an arbitrary field in a mixture of gases.

By bridging rigorous kinetic theory with accessible computation, IMoS 2 is not merely an incremental improvement over MOBCAL or earlier IMoS versions. It is a platform for the next generation of IMS research: one capable of addressing both established CCS benchmarks and emerging questions at the limits of IMS resolution. IMoS 2 resolves the stochastic phase-space evolution of ions under external force, from which macroscopic transport properties emerge as ensemble-averaged quantities. With its public release, we aim to provide the community with a transparent, extensible, and reproducible tool that advances the fundamental theory and practical utility of ion mobility.

DISCUSSION SECTION

The Generalized Boltzmann Equation (WCU/WUB) to Be Solved

While it can be further relaxed, let us consider a generalized Boltzmann kinetic equation in an extended phase space including translational, orientational, and rotational degrees of freedom, combining Curtiss-type rotational transport with a collision operator consistent with Wang−Chang−Uhlenbeck detailed balance.[61−64] For simplicity, let us assume we only have a single gas and that the ion's density is small enough so that two ions will not interact with each other. Without being extremely rigorous, the WCU equation for a distribution of an ion of mass m and charge q, $f(\mathbf{r}, \mathbf{v}, \mathcal{R}, \mathbf{J}, t)$, may be written as

$$\frac{\partial f(\mathbf{r}, \mathbf{v}, \mathcal{R}, \mathbf{J}, t)}{\partial t} + \mathbf{v}\cdot\nabla_{\mathbf{r}} f + \frac{q}{m}\mathbf{E}\cdot\nabla_{\mathbf{v}} f + \Omega(\mathcal{R}, \mathbf{J})\cdot\nabla_{\mathcal{R}} f + \mathbf{N}(\mathbf{r}, \mathcal{R}, t)\cdot\nabla_{\mathbf{J}} f = \sum_{s=1}^{S} C_s[f] \quad (1)$$

With the collision term being:

$$C_s[f] = n_s \sum_l w_{sl}(T_s) \int_{R^3} \mathrm{d}\mathbf{u}_s F_s(\mathbf{u}_s) \int \mathrm{d}\Gamma\; g_s[f'(\mathbf{r}, \mathbf{v}', \mathcal{R}', \mathbf{J}', t) - f(\mathbf{r}, \mathbf{v}, \mathcal{R}, \mathbf{J}, t)]\mathcal{K}_{sl}(\Gamma; \mathbf{v}, \mathcal{R}, \mathbf{J}, \mathbf{u}_s) \quad (2)$$

The ion velocity distribution is initially a function of position $\mathbf{r}$, translational velocity $\mathbf{v}$, orientation $\mathcal{R}$, angular momentum $\mathbf{J}$ and time. The first three terms on the left-hand side of eq 1 correspond to the typical advection from the Boltzmann equation[65] subject to an electric field $\mathbf{E}$ while the remaining two terms correspond to advection terms in a 12-dimensional space that emerge from the kinematic generator of rigid-body rotation $\boldsymbol{\Omega}$ (dependent on the inertia tensor) and the dipole moment torque $\mathbf{N}$. The right-hand side of eq 1 is expanded on eq 2 and corresponds to the collision term for a series of neutral species s of gas number density n_s and temperature T_s and velocity $\mathbf{u}_s$. Apparent in this term are weights w_{sl} corresponding to partition functions of internal states l with degeneracy g_{sl}:

$$w_{sl}(T_s) = \frac{g_{sl} e^{-\frac{\epsilon_{sl}}{k_b T_s}}}{Q_{s,\mathrm{int}}(T_s)}; \quad Q_{s,\mathrm{int}} = \sum_l g_{sl} e^{-\frac{\epsilon_{sl}}{k_b T_s}} \quad (3)$$

In eq 2 g_s is the relative velocity while $f(\mathbf{r}, \mathbf{v}', \mathcal{R}', \mathbf{J}', t) - f(\mathbf{r}, \mathbf{v}, \mathcal{R}, \mathbf{J}, t)$ represents the replenishment (coming from prime sources) and extinguishment of ions of a given class of infinitesimal states in the extended space. $d\Gamma$ denotes the geometric measure over collision configurations (not only impact parameters but now relative orientation) parametrizing how encounters are sampled. $\mathcal{K}_{sl}$ is a collision kernel that accounts for probabilistic or averaged interaction physics associated with unresolved internal degrees of freedom (e.g., deformation, vibrational excitation, or state-to-state

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transitions) and reduces to unity in this manuscript since only rotation is included herein. Finally, the gas velocity distribution is modeled from a Maxwell−Boltzmann equilibrium distribution and given by[66−68]

$$F_s(\mathbf{u}_s) = \left(\frac{m_s}{2\pi k_B T_s}\right)^{3/2} e^{\left(-\frac{m_s|\mathbf{u}_s|^2}{2k_B T_s}\right)} \tag{4}$$

This natural extension of the Boltzmann equation extensively complicates the collision term making it analytically intractable for anything but the simplest cases. Direct solutions of the resulting equation using conventional quadrature or discretization methods is prohibitively cumbersome due to the high dimensionality of the extended phase space. This realization leads to the conclusion that a numerical approach is necessary. IMoS 2 provides a quantitative solution strategy by bypassing the explicit evaluation of the high-dimensional collision operator altogether. The method indirectly solves the kinetic equation through a trajectory-resolved molecular-dynamics framework, in which an ion is propagated within an explicit local environment of neutral gas molecules and evolves deterministically under the combined influence of external fields and ion−neutral interaction forces. In what follows, the software will be described in detail.

IMoS 2 Description

The guiding principle of the method is to reproduce the assumptions underlying the Boltzmann equation as faithfully as possible, while retaining the flexibility to relax or modify those assumptions when required. To this end, the algorithm has been developed entirely from first-principles, with the explicit objective of recovering known Boltzmann-equation quadratures and limiting behaviors under idealized conditions. This design ensures consistency with established kinetic theory in regimes where analytical results exist, while providing a controlled framework for systematic extensions beyond those limits.

The Domain, the Ion and the Gas

The IMoS 2 simulation domain is defined as an NVT ensemble of an ideal neutral gas, in which gas-phase constituents do not interact with one another. The domain is constructed to accommodate an ion propagating under the influence of a prescribed electric field and is chosen to be sufficiently large such that boundary effects do not interfere with ion−neutral interactions. In practice, the domain dimensions are several times larger than the effective interaction region of the ion, with an elongated geometry in the direction of the applied field. By default, the simulation box contains 800 gas molecules, and its size is a rectangle of $3l \times l \times l$, where l is a function of the pressure and temperature chosen for the gas, although these values are adjustable.

The polyatomic ion is modeled as a rigid rotating body, whose optimized geometry is obtained from density functional theory (DFT) calculations. Partial atomic charges and the permanent dipole moment are explicitly included, and the ion's center of mass and moment of inertia tensor are computed directly from the optimized structure. The ion undergoes both translational acceleration and rotational torque in response to an externally applied electric field $\mathbf{E}(x,t)$ which may be a function of space and time. Ion−neutral interactions are described using a combination of ion-induced dipole forces and short-range repulsion−dispersion interactions represented by a 4−6−12 potential. A sphere of influence establishes an interaction cutoff where the radius is given by the ratio of potential energy to kinetic energy of the gas. As a default, this value is considered to be PE/KE = 0.5%, but it is modifiable. Unlike previous mobility algorithms which only allow a single gas molecule to interact with the ion, any number of neutral gas molecules may enter the sphere of influence and interact with the ion, effectively avoiding artificial restrictions on collisions multiplicity. The polarizability of the gas and the L−J parameters are tabulated and can be modified and optimized for the different gases. While the Buckingham potential might be a more optimal choice for higher energy interaction,[24] it is not implemented in IMoS 2 due to its computational cost.

The neutral gas is initially modeled as a single, globular interaction center, even for diatomic or polyatomic species. This choice is motivated both by computational convenience and by established convention in kinetic theory and collision cross section calculations. Because ion−neutral interactions are treated explicitly at the level of interaction energies, the neutral gas molecules can be represented using a reduced description consisting of a point particle interacting through a single effective pair potential. This approximation has been widely employed in previous collision cross section quadrature approaches and is retained here for consistency. Gas mixtures are readily incorporated by specifying the appropriate molecular fractions for each neutral species. Neutral gas translational velocities are sampled from a Maxwell−Boltzmann distribution at the temperature relevant to the experiment, consistent with the equilibrium assumption underlying the Boltzmann equation and preserved in the present framework.[69]

The Marching Algorithm and Timesteps

The algorithm employs multiple adaptive time steps to optimize computational efficiency while maintaining numerical accuracy. It alternates between an event-driven analytical propagation regime and a time-driven molecular-dynamics regime. When the ion is not interacting with any neutral gas molecule, its motion is advanced using an outer propagation scheme in which the time step is determined by an analytical marching procedure that computes the time to the next possible ion−neutral or neutral-boundary encounter. In this regime, the relative motion is governed by central forces, allowing the collision time to be obtained from a closed-form solution. When an interaction between ion and neutral is detected, the system is advanced precisely to the onset of the collision, and control is transferred to an inner integration routine that resolves the interaction dynamics within the sphere of influence. During this phase, the outer propagation scheme remains active to account for the possibility that additional gas molecules may enter the sphere while an interaction is already in progress.

Within the interaction region, the coupled translational and rotational dynamics of the ion and nearby gas molecules are integrated using a Beeman algorithm for translational motion and a predictor−corrector rotational Verlet scheme for rotational degrees of freedom.[70−72] These integrations are performed with a substantially smaller time step, typically on the order of 1 fs, though this value is adjustable. The Beeman integrator is selected for its improved velocity accuracy leading to more stable energy conservation behavior relative to the velocity Verlet method at a modest increase in computational cost, while the rotational Verlet scheme is used in conjunction with a quaternion-based representation of orientation to prevent spurious energy loss and numerical drift. The inner integration continues until all gas molecules have exited the ion's sphere of influence, at which point the algorithm reverts to the outer

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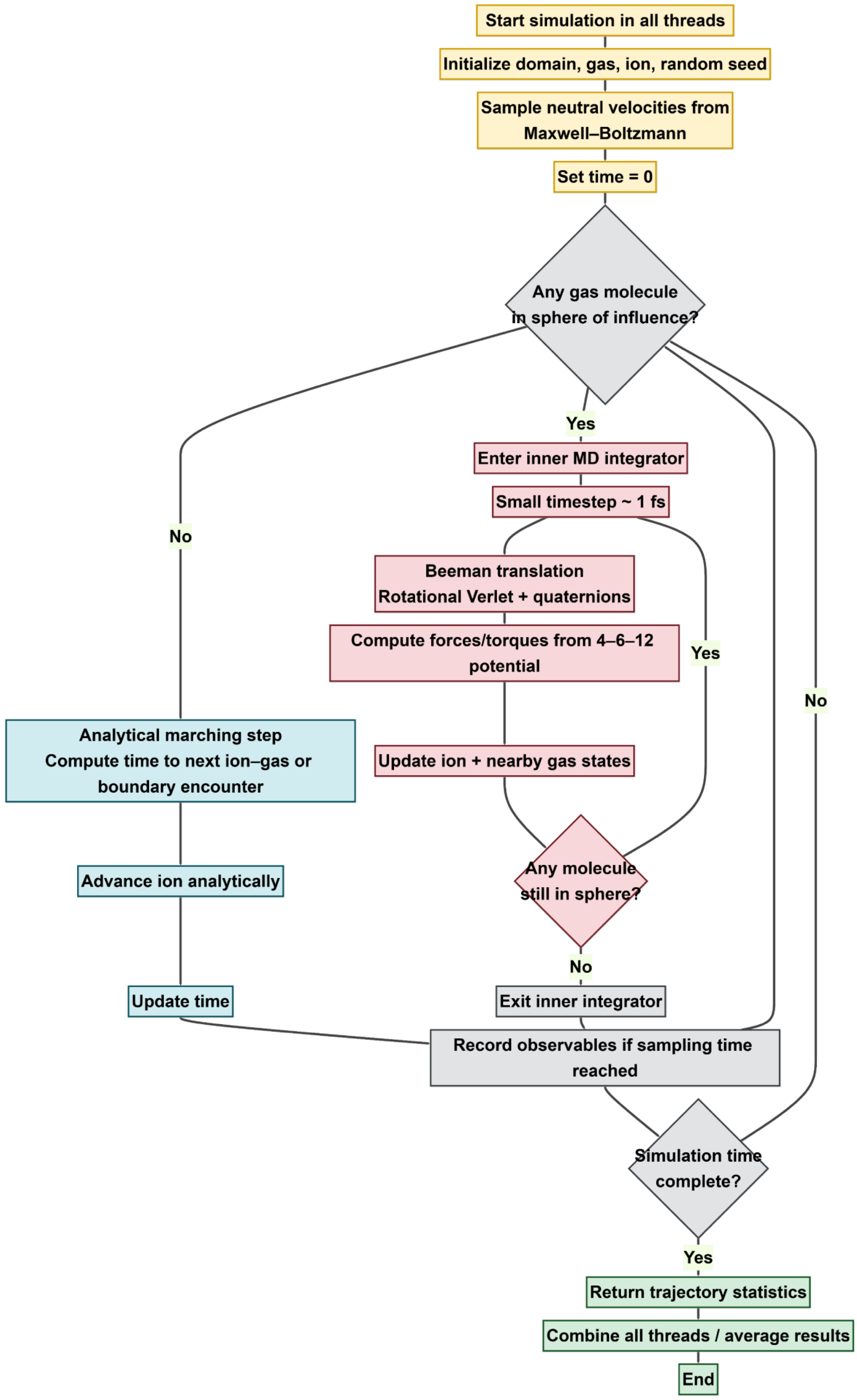

Figure 1. Flowchart of the IMoS 2 hybrid Monte Carlo−molecular dynamics workflow. The ion alternates between analytical free flight to the next encounter and an inner MD integrator that explicitly resolves short-range ion−gas collisions within a sphere of influence using subfemtosecond timesteps and 4−6−12 interaction forces.

marching scheme and resumes analytical propagation until the next interaction occurs. A flowchart of the process is given in Figure 1.

Modularity and Parallelization

IMoS 2 is designed with a modular architecture that facilitates systematic modification and extension of the underlying physical description without requiring changes to the overall algorithmic structure. Key components, such as the size of the ion's sphere of influence and the manner in which interactions are resolved within that region, can be readily adjusted. For example, the current 4−6−12 interaction potential may be straightforwardly replaced by elastic or inelastic hard-sphere scattering descriptions, providing a clear pathway toward incorporating vibrational degrees of freedom and more detailed molecular dynamics treatments in future implementations. The algorithm is parallelized using OpenMP, allowing multiple statistically independent realizations of ion trajectories to be propagated concurrently using different random seeds. Provided that each

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individual trajectory ensemble satisfies the appropriate statistical requirements, results from these independent simulations can be combined to improve statistical convergence and reduce uncertainty in the computed transport properties. A schematic representation of the simulation domain and ion geometry is shown in Figure 2a, while Figure 2b illustrates ion−neutral interactions within the defined sphere of influence.

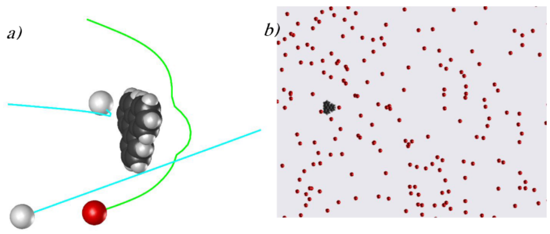

Figure 2. (a) Representative snapshot of the simulation domain showing the ion (dark cluster) immersed in a bath of neutral gas molecules sampled from a Maxwell−Boltzmann distribution. (b) Close-up of a collision event within the sphere of influence, where nearby neutrals are propagated with the inner MD integrator while the ion trajectory is updated explicitly.

■ RESULTS

The range of results obtainable from the algorithm is broad and depends on the specific quantities of interest. In the following, we focus on the most commonly extracted observables and compare them directly with results obtained from quadrature-based approaches used in previous methods. During propagation, instantaneous quantities such as ion position, translational velocity, and angular velocity throughout the simulation are recorded at a user selected fixed recording interval. The volume of raw data required to generate statistically significant results is substantial, typically in the order of several gigabytes, but the resulting statistics are correspondingly high fidelity. A representative example of time-resolved trajectories is shown in Figure 3. Note that the changes in direction in Figure 3 are not necessarily from collisions, but rather each direction is an average from all collisions during the cadence of choice. The results from the path and velocity can be used to observe the difference between true and average mobility, as the ion covers substantially more distance than suggested by instrument calculations.[32] The velocity and angular velocity distribution functions of the ions can also be reconstructed from the data acquired, thereby providing a direct numerical solution to the Boltzmann/WCU equation. Examples of these distributions are presented in Figure 4 for the X, Y and Z directions. The same analysis framework can be applied to electric fields that vary in both space and time, such as those from a T-wave or FAIMS.[73−76] An example is shown in Figure 5, where an electric field of the form $E = A\sin(kx + \{\backslash omega\}t)$ is used to reproduce the electric field experienced by an ion over approximately 8000 cycles. The electric field shown mimics the results produced by SIMION simulations of SLIM devices, further validating ots results.[77]

In addition to distribution-level observables, the algorithm records the mean ion drift velocity, which can be used to infer ion mobility, particularly under constant-field conditions. Drift velocities are computed independently for each parallel execution thread, enabling direct assessment of statistical convergence across cores. As the simulation length increases, the variance between independently sampled trajectory ensembles decreases accordingly. As a representative example, Figure 6a shows results for the ion triphenylene obtained using 256 cores, each accumulating approximately 23 million collisions, of the number of total collisions versus mean ion drift velocity results per core. The comparatively large number of collisions required to achieve statistical convergence, relative to methods such as MobCal or earlier versions of IMoS (which typically require on the order of 10^6 collisions), arises from the inability to apply Chapman−Enskog linearization to the exponential form of the integral quadrature inherent to the present formulation. This limitation is intrinsic to the method and reflects a deliberate trade-off required to retain continuity effects and full dynamical resolution. Physically speaking, the momentum transfer from collisions happening from the front and back of the ion cancel each other out which can analytically

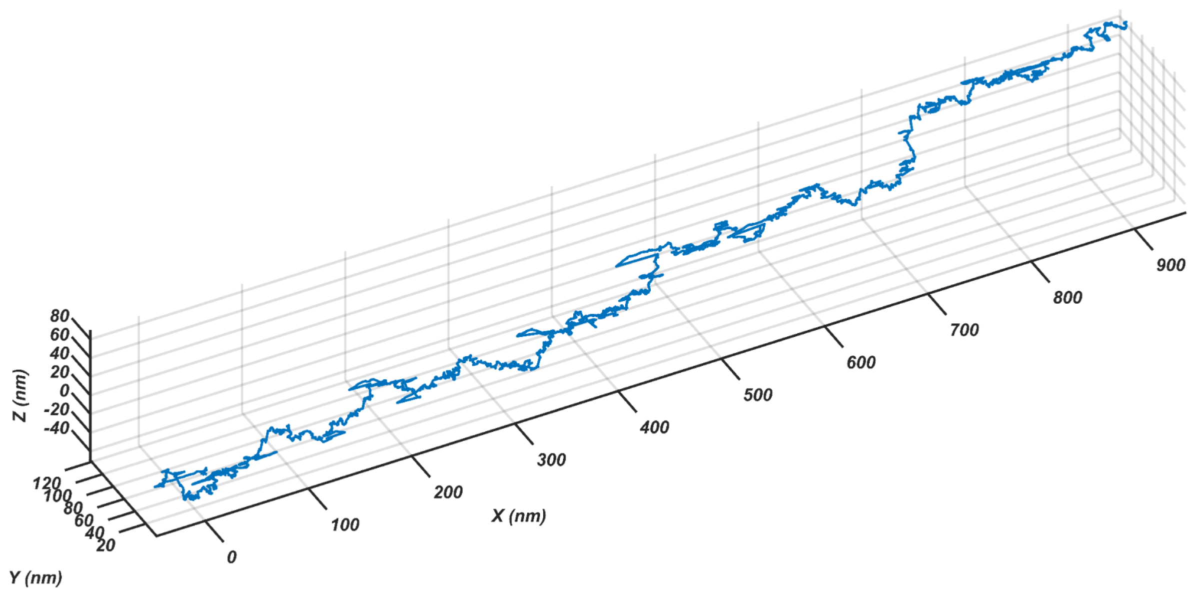

Figure 3. Three-dimensional ion trajectory computed with IMoS 2. Ion positions are sampled every 1 ns.

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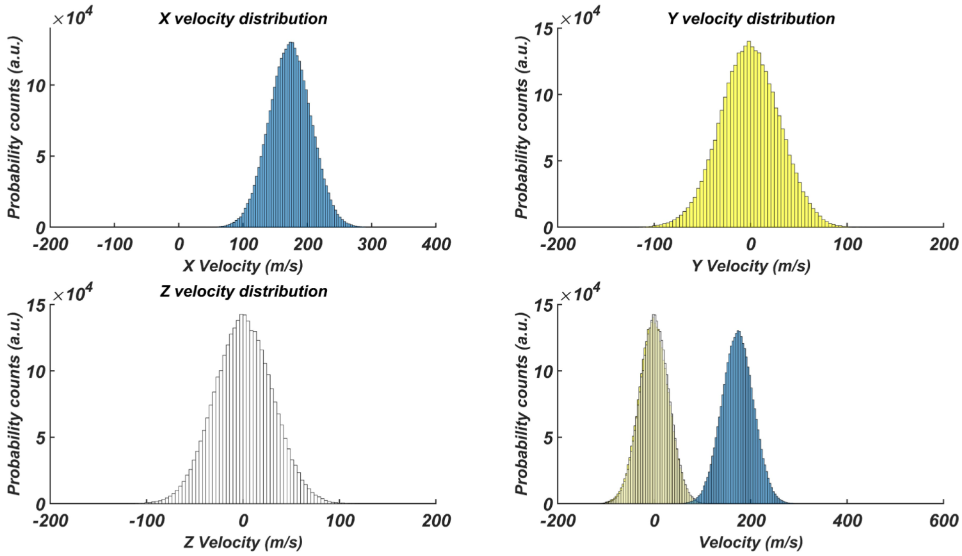

Figure 4. Velocity distributions extracted from IMoS 2 trajectories. Histograms of the Cartesian ion velocity components show thermalized transverse motion (Y and Z) and a field-induced drift along X. The bottom-right panel compares the equilibrium thermal distribution with the shifted drift distribution along the transport direction.

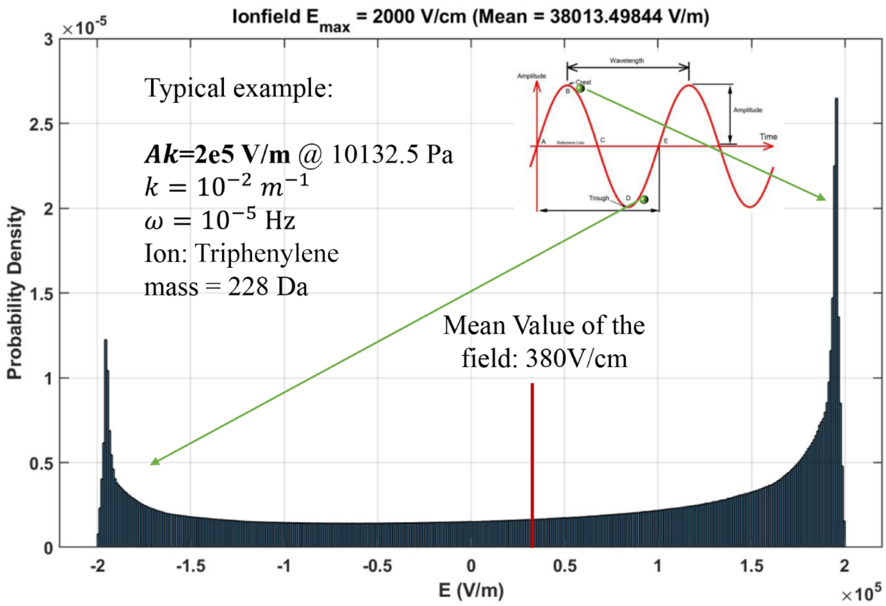

Figure 5. Probability density of the instantaneous electric field sampled along an ion trajectory in a traveling-wave (T-wave) field, modeled as $E = A\sin(kx + wt)$. Results correspond to approximately 8000 cycles for the representative conditions shown, illustrating the nonuniform field exposure experienced by the ion during transport.

be removed with the linearization of the equation assuming all collisions are independent of each other, an assumption not valid here. The idea associated with this process is further illustrated schematically in Figure 6b and a theoretical explanation of the reason is given by eq (5a) in ref 28.[28,35]

IMoS 2 Results and Validation

The first test of the new algorithm would be to compare the result to the existing quadrature methods at low/negligible fields. The first moment of the Boltzmann equation, assuming a negligible and constant electric field, yields the well-regarded Mason Schamp equation which is given by

$$K_0 = \frac{3e}{16}\sqrt{\frac{2\pi}{\mu kT}}\frac{1}{\Omega(1,\,1)(T_{\text{eff}})} \tag{5}$$

Where the Collision Cross Section (CCS) is given by the quadrature:

$$\Omega(l,\,n) = \int e^{-g^{*2}} g^{*2n+3} Q^{(l)} dg^* \tag{6}$$

$$Q^{(l)} = \int 2\pi(1 - \cos^{l}(\chi))bdb \tag{7}$$

Typical quadrature methods, e.g., Mobcal, initially use Monte Carlo simulations to calculate the deflection angle χ and then numerically calculate the integrals. In the case of IMoS 1, instead of the calculation of the integral quadratures, a momentum transfer approach is employed, which is considered to be equivalent. The mobility can also be calculated from IMoS 2

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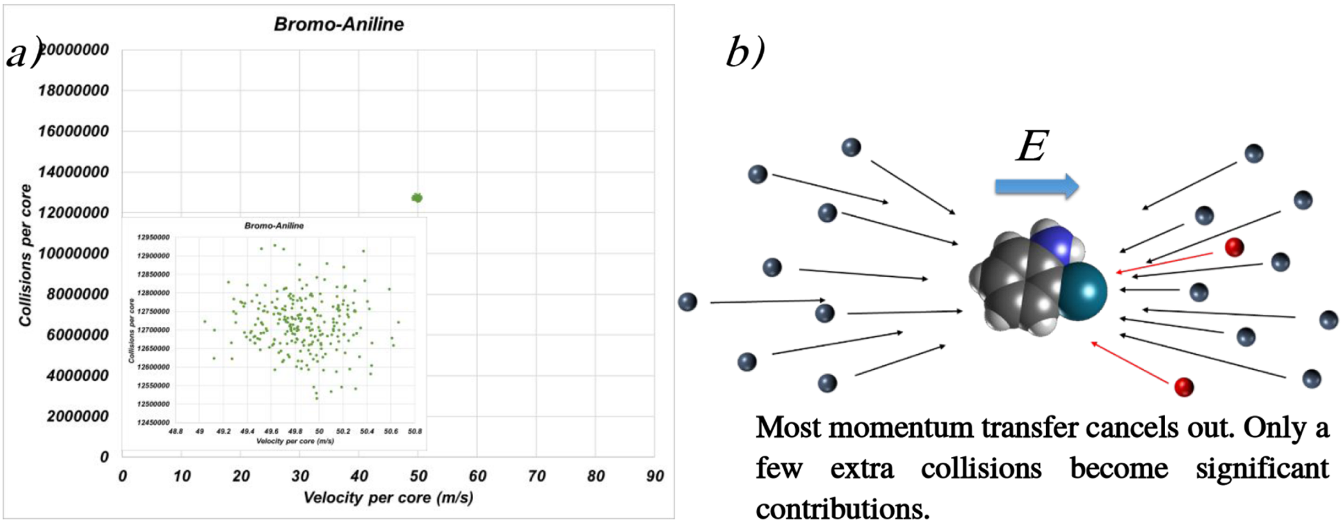

Figure 6. (a) Mean ion drift velocity versus total number of ion−gas collisions accumulated per core for triphenylene, with each point corresponding to an independent parallel thread (256 cores). (inset) Zoomed view highlighting the tight clustering of independently sampled cores after sufficient trajectory length. (b) Schematic illustration of stochastic momentum transfer from front- and back-side collisions that largely cancel (gray-blue cancel each other), necessitating extensive sampling for statistical convergence.

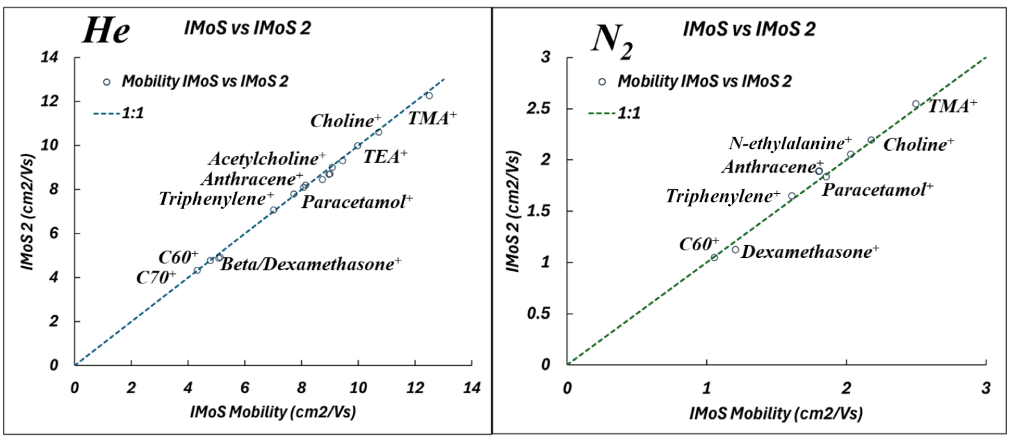

Figure 7. Comparison of low-field ion mobilities predicted by IMoS 2 and IMoS 1 in (left) He and (right) N_2. Mobilities from IMoS 2 are obtained from the drift $\frac{v_d}{\mathbf{E}}$ under constant, negligible-field conditions, while IMoS 1 values are computed using the traditional quadrature-based momentum-transfer framework. The dashed line indicates 1:1 agreement.

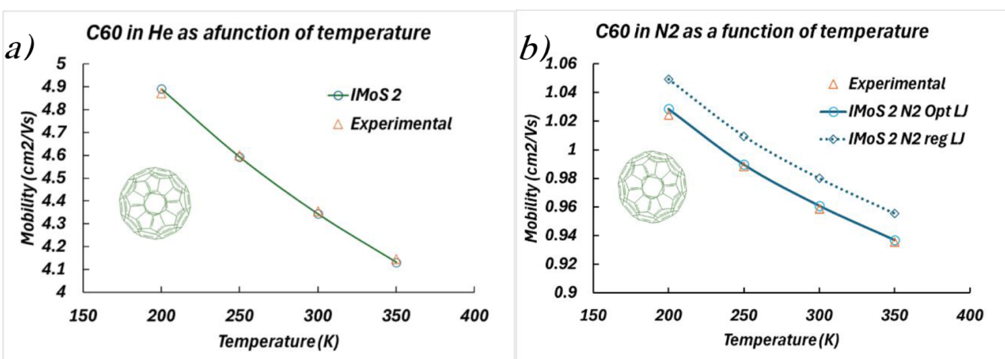

Figure 8. Temperature dependence of C60$^+$ mobility predicted by IMoS 2 compared with experimental measurements in (a) He and (b) N_2.

from the average drift velocity by using the equation $v_d = KE$, assuming constant field and no relaxation.

The comparison results are shown in Figure 7 where $\frac{v_d}{\mathbf{E}}$ calculated from IMoS 2 at low fields is compared to IMoS 1 results in both N_2 and He. Despite the difference in approach, the results are remarkably close. Some of the differences between the two results can be attributed to rotational and multicollision effects, which are not taken into account in the quadrature methods.

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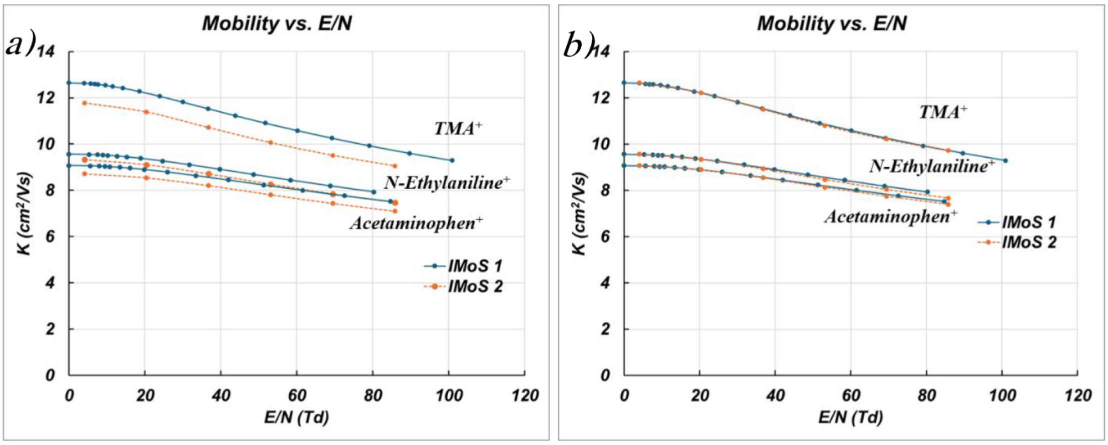

Figure 9. Field-dependent mobility K as a function of reduced field $\frac{E}{N}$ predicted by IMoS 2 compared with IMoS 1 for representative ions. (a) Direct comparison showing close agreement across the full field range. (b) After correcting the small zero-field offset, the results are nearly indistinguishable.

We must emphasize that IMoS 2 calculates the velocity distribution and then calculates the first moment which agrees with the analytical quadrature, further establishing that the method calculates the full Boltzmann/WCU equation.

The variation of mobility with temperature can also be tested against experimental results.[22] Figure 8 compares IMoS 2 predictions in a) He and b) N_2 for $C60^+$ as a function of temperature with experimental results for the same temperature set. Agreement in He is excellent across all temperatures. In N_2, the correct trend is reproduced with small quantitative deviations attributable to the optimized L−J parameters for carbon, which were tuned to balance accuracy across a broader set of organic species. Similar discrepancies are observed in IMoS 1 and can be removed with minor L−J parameter adjustments. Once again, IMoS 2 is matching the result of the quadrature quantitatively.

An ultimate validation of IMoS 2 as a direct solver of the WCU equation is obtained by testing the algorithm under arbitrary electric field profiles for light monatomic gases. Experimental data sets in this regime are scarce; therefore, benchmarking is performed against predictions from the two-temperature theory and conventional collision-integral quadratures. The comparison, shown in Figure 9a, demonstrates very close agreement. After correcting the small offset observed at zero field, the results become nearly indistinguishable, as illustrated in Figure 9b.

For heavier gases such as N_2, IMoS 2 continues to accurately predict the field-dependent mobility in the low $\frac{E}{N}$ regime, as shown in Figure 10. At higher fields, however, systematic deviations from the two-temperature theory emerge. These discrepancies are consistent with the appearance of inelastic collisions that transfer energy into internal rotational (and vibrational, although not explored here) degrees of freedom, thereby violating the assumptions underlying the two-temperature approximation. This behavior was anticipated in the classical treatment of Mason and McDaniel, and is demonstrated here computationally for the first time.

While not shown in this work, the algorithm has already shown that it is capable of differentiating shifts in Moment of Inertia and Center of Mass and the rotational effects have been confirmed both experimentally and through quadratures.[40,57,58] The algorithm can also include the effect of the dipole moment as previously shown. While this effect has not been tested here for the lack of experiments or theory that would properly test it, there is no reason to indicate that the effect cannot produce significant shifts on appropriate ions and fields. Finally, mixtures of gases can be included as well but secondary effects and departures from Blanc's law have yet to be tested.

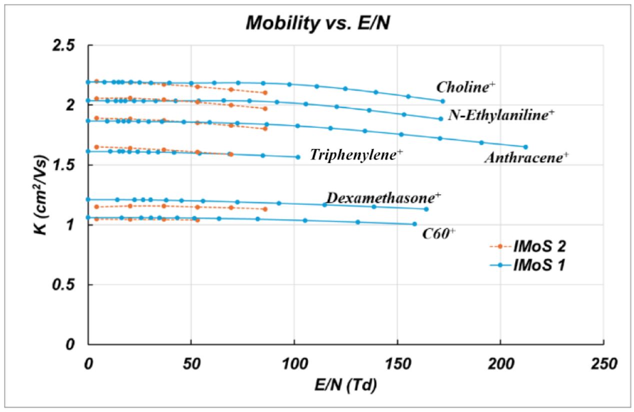

Figure 10. Field-dependent mobility K as a function of reduced field $\frac{E}{N}$ in N_2 for representative ions, comparing IMoS 2 with IMoS 1 (two-temperature theory). Excellent agreement is observed in the low-field regime. At higher fields, systematic deviations emerge as inelastic energy transfer into internal rotational (and vibrational) modes violates the assumptions of the two-temperature approximation.

■ CONCLUSIONS

In this work, we introduced IMoS 2, a trajectory-based, first-principles solver for the Wang Chang−Uhlenbeck (WCU) kinetic equation that directly simulates ion−neutral collisions under arbitrary electric field conditions. By replacing traditional collision integral quadratures and momentum transfer approximations with explicit Monte Carlo molecular dynamics, the method provides a physically transparent and numerically robust route to compute transport properties without relying on low-field or near-equilibrium assumptions. Across all regimes tested, IMoS 2 reproduces classical transport theory where it is valid and naturally extends beyond it when those assumptions break down, thereby establishing a unified and predictive framework for ion mobility calculations. The algorithm is freely available from the corresponding author's website, www.

imospedia.com. The following accomplishments are highlighted:

- Developed a full trajectory level Monte Carlo solver for the WCU equation that eliminates the need for precomputed collision integrals or empirical momentum transfer models.
- Because the ion retains memory of previous collisions anyeffects such as heating, rotational/vibrational energy exchanges, collision induced dissociation, inelastic energy transfer, reduced mass considerations and secondary effects in gas mixtures are considered in the calculations.
- The algorithm provides angular and translational velocity distributions as well as ion trajectories. The field employed can be a function of space and time so nonlinear and relaxation effects from T-waves, FAIMS and RF can also be studied.
- Demonstrated quantitative agreement with IMoS 1 and regular quadrature methods in the low field limit for multiple buffer gases, confirming equivalence with established momentum transfer formalisms. It has also been tested for different temperatures successfully.
- Validated the algorithm against two-temperature theory and collision integral quadratures for light monatomic gases under arbitrary electric fields, showing near-exact agreement after zero field corrections.
- Accurately predicted field-dependent mobility trends for heavier gases and identified systematic deviations from two temperature theory at elevated $\frac{\mathbf{E}}{\mathbf{N}}$, attributable to inelastic energy transfer into internal degrees of freedom.
- Provided computational evidence of rotational and multicollision effects and the breakdown of simplified transport closures, phenomena previously anticipated theoretically but not explicitly resolved in simulations.
- Established IMoS 2 as a general-purpose platform capable of treating complex potentials, nonuniform or time varying fields, and strongly nonequilibrium conditions relevant to modern high resolution ion mobility instrumentation.

Future developments will focus on extending IMoS 2 toward increasingly realistic and experimentally relevant systems. Planned efforts include incorporation of detailed explicit treatment of polyatomic internal energy exchange (including vibrations), and coupling to spatially varying electric field maps representative of modern drift tube, SLIM, and varying field drift tube architectures, including RF. Algorithmic acceleration and parallelization will enable larger ensembles and longer time scales, facilitating direct simulation at low pressures and larger molecular systems. Together, these advances will position IMoS 2 as a predictive design tool for next generation ion mobility instruments, energy exchange prediction and reaction in the gas phase and as a general computational framework for non-equilibrium gas phase transport phenomena.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.6c01278.

The manual of the algorithm has been added as Supporting Information with suitable examples of calculations (PDF)

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■ ABBREVIATIONS

IMoS, Ion Mobility Software Suite; IMoS 1, First-generation; IMoS 2, Second-generation; IMS, Ion mobility spectrometry; MS, Mass spectrometry; WCU, Wang−Chang−Uhlenbeck equation; WUB, Wang−Chang−Uhlenbeck−de Boer equation; MD, Molecular dynamics; MC, Monte Carlo; MC−MD, Hybrid Monte Carlo−molecular dynamics; CCS, Collision cross section; L−J, Lennard−Jones; SLIM, Structures for lossless ion manipulation; FAIMS, High-field asymmetric waveform ion mobility spectrometry; T-wave, Traveling wave; E/N, Reduced electric field (electric field divided by gas number density); K, Ion mobility; K_0, Reduced (standard) ion mobility; v_d, Ion drift velocity; Ω, Collision integral; He, Helium; N_2, Nitrogen; TMA, Tetramethylammonium; TEA, Tetraethylammonium

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